



SVP-25-016

10 CFR 50.73

March 27, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1 and 2
Renewed Facility Operating License No. DPR-29 and DPR-30
NRC Docket No. 50-254 and 50-265

Subject: Licensee Event Report 254/2025-002-00 "Control Room Emergency Ventilation Air Conditioning System Inoperable due to Logic Issue"

Enclosed is Licensee Event Report 254/2025-002-00 "Control Room Emergency Ventilation Air Conditioning System Inoperable due to Logic Issue," for Quad Cities Nuclear Power Station, Units 1 and 2.

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D) for any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Conner Bealer at 779-231-6207.

Respectfully,

A handwritten signature in black ink, appearing to read "Doug Hild", written over a light blue horizontal line.

Doug Hild
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Quad Cities Nuclear Power Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00254

3. Page

1 OF 4

4. Title

Control Room Emergency Ventilation Air Conditioning System Inoperable Due to a Logic Issue

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
01	26	2025	2025	- 002 -	00	03	27	2025	Quad Cities Unit 2	☒ 050	00265
									Facility Name N/A	☐ 052	Docket Number N/A

9. Operating Mode

1 – Power Operation

10. Power Level

091%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Richard Swart – Regulatory Assurance

Phone Number (Include area code)

309-227-2810

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	VI	52	C770	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 01/26/2025 at 14:50, the Control Room Emergency Ventilation (CREV) Air Conditioning (AC) system was declared inoperable due to a trip of the Refrigeration Condensing Unit (RCU) feed breaker. Technical Specification (TS) 3.7.5 Condition A was entered at that time. Control Room operators discovered that the indicating light for the RCU was not lit. An operator was dispatched and found that the feed breaker had tripped. After post maintenance testing of the breaker, TS 3.7.5 Condition A was exited on 02/07/2025 at 00:30.

Initial troubleshooting narrowed down to two potential causes which includes breaker failure and loose breaker wiring. The causal product identified a design vulnerability where a simultaneous failure of a pressure switch and time delay relay could also cause the issue. The first two causes were addressed by breaker inspection and overhaul. The final cause will be addressed by replacement of a time delay relay and low suction pressure switch.

This is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D), any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Quad Cities Nuclear Power Station Unit 1	☒ 050	2. DOCKET NUMBER 00254	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 002	REV NO. - 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric – Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Control Room Emergency Ventilation Air Conditioning System Inoperable Due to a Logic Issue

A. CONDITION PRIOR TO EVENT

Unit: 1	Event Date: January 26, 2025	Event Time: 1450 hours CST
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 91%

No systems, structures, or components that were inoperable at the start of the event contributed to the event.

B. DESCRIPTION OF EVENT

On January 26, 2025, at 14:50, Operations identified that there was no indicating light for the Control Room Emergency Ventilation (CREV)[VI] Refrigeration [RFU] Condensing Unit (RCU) 0-9400-102 on the Main Control Room (MCR)[NA] Panel 912-5. An operator was dispatched and found that local indications also showed the RCU to have no power and the breaker [BKR] in a tripped condition. The station entered Technical Specification (TS) 3.7.5 Condition A for an inoperable CREV Air Conditioning (AC) [ACU] system. ENS Notification #57515 was made in accordance with 10 CFR 50.72(b)(3)(v)(D) for any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

The CREV RCU is known to start with a high inrush of current. This was previously attributed to a "double start" condition where it would start once and then within seconds start again. Between 2009 and 2012, numerous trips of the CREV RCU breaker occurred. To eliminate these events a modification was installed in 2009 to add a time delay relay to the low suction pressure switch. The modification was to eliminate/reduce the number of "double starts" on the compressor and reduce the breaker tripping. In addition, the breaker dial setting was changed in 2012 from 8.1 to 9.4 to provide additional margin in the breaker trip setting.

All these previous events and changes to the logic and breaker settings have made the system more robust over time. Between 2012 and 2020, and again after 2020, there were no CREV RCU breaker trip events.

Because the CREV AC system is a single train system, this report is submitted in accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), which requires the reporting of an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

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NARRATIVE**C. CAUSE OF EVENT**

Initial troubleshooting narrowed down to two potential causes which includes breaker failure and loose breaker wiring. The causal product identified a design vulnerability where a simultaneous failure of a pressure switch and time delay relay could also cause the issue. A simultaneous failure could have potentially allowed a high inrush starting current during one of the CREV RCU compressor pump down cycles. A pump down cycle happens approximately every four hours. During the pump down cycle the RCU compressor starts for a short period to pump any liquid refrigerant from the suction side of the compressor to prevent compressor damage. The first two causes were addressed by breaker inspection and overhaul. The final cause will be addressed by replacement of a time delay relay [62] and low suction pressure switch [PS].

D. SAFETY ANALYSISSystem Design

The purpose of the Control Room Emergency Ventilation system is to maintain the proper air environment for instrumentation and personnel in the Main Control Room. The CREV AC system is a single train safety-related system that is designed to maintain design temperature in the Control Room Envelope (CRE) under post-accident conditions. The loss of CREV AC could impact the plant's ability to mitigate the consequences of an accident. The CREV RCU is a key component of the CREV AC system and is a Carrier reciprocating compressor with a capacity of 90 tons. The backup to the CREV system is the Train A Control Room HVAC system, which is a non-safety related system.

Per Updated Final Safety Analysis Report (UFSAR) Section 6.4, "Habitability Systems," the Control Room Heating, Ventilation and Air conditioning (HVAC) systems are capable of maintaining the control room atmosphere suitable for occupancy throughout the duration of a Design Basis Accident (DBA). The RCU provides cooling to support CREVs operation. The emergency operation of CREVs requires manual actions.

Safety Impact

The safety significance of this event was minimal. The CREV system filtration capability was not lost, and control room temperature was maintained during this event. In addition, the non-safety related Train A Control Room HVAC system was available and operating throughout this event.

E. CORRECTIVE ACTIONS

Immediate: The breaker cubicle was inspected and overhauled.

Follow-up: The time delay relay and the low suction pressure switch will be replaced.

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NARRATIVE**F. PREVIOUS OCCURENCES**

The station events database, LERs and IRIS were reviewed for similar events at Quad Cities Nuclear Power Station. This event was a CREV RCU power issue, which was caused by a logic issue.

LER (254/2020-001-00) Control Room Emergency Ventilation Air Conditioning System Inoperable Due to Incorrect Breaker Setting, 04/14/2020. The CREV RCU was declared inoperable when the electrical feed breaker to the RCU was found in a tripped condition. This event was caused by an incorrect breaker setting for the RCU feed breaker. The breaker setting was changed in 2012, however the guidance documents for the preventative maintenance tasks on the RCU feed breaker were not updated prior to the PM task completion in 2013.

G. COMPONENT FAILURE DATA

Failed Equipment: BREAKER, MCC; 480V
Component Manufacturer: Cutler-Hammer/Eaton
Component Model Number: KD3250
Component Part Number: N/A

This event has been reported to IRIS.